2 **(a)** State the principle of conservation of momentum.

....

....

....

.... [2]

(b) A stationary boat of mass 160 kg floats on water. The mast of the boat is vertical. A sack of mass 23 kg is thrown at the boat, as shown in Figure 2.1.

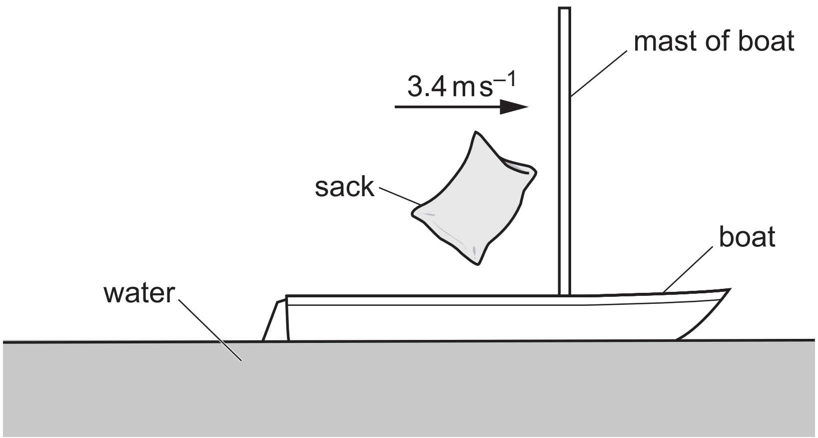

Figure 2.1

Immediately before colliding with the mast of the boat, the sack is travelling horizontally with a speed of $3.4\,\mathrm{m\,s^{-1}}$.

The sack collides with the mast and remains in contact with it, without rebounding.

Assume that resistive forces acting on the boat during the collision are negligible.

(i) Calculate the horizontal speed v of the sack and boat immediately after the collision.

v = ... $\mathrm{m\,s^{-1}}$ [2]

(ii) The total kinetic energy of the sack and boat immediately before the collision is E_1. The total kinetic energy of the sack and boat immediately after the collision is E_2.

Determine the ratio $\frac{E_2}{E_1}$.

ratio = ... [3]

(iii) State and explain whether the collision between the sack and boat is elastic or inelastic.

....

....

.... [1]

(c) After a short time, the boat in **2(b)** is at rest again.

(i) Show that the upthrust acting on the boat with the sack is 1800 N.

[1]

(ii) The boat displaces water with a volume of $1.80 \times 10^5\,\text{cm}^3$.

Calculate the density of the water. Give your answer to 3 significant figures.

density = ... $\text{kg}\,\text{m}^{-3}$ [2]

[Total: 11]